Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void quickSort(int a[], int low, int high)
{
    int i = low, j = high;
    int pivot = a[(low + high) / 2];
    int temp;

    while (i <= j)
    {
        while (a[i] < pivot)
            i++;

        while (a[j] > pivot)
            j--;

        if (i <= j)
        {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
            i++;
            j--;
        }
    }

    if (low < j)
        quickSort(a, low, j);

    if (i < high)
        quickSort(a, i, high);
}

int main()
{
    int n, i;

    scanf("%d", &n);

    int a[n];

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    quickSort(a, 0, n - 1);

```

```
47      for (i = 0; i < n; i++)
48          printf("%d ", a[i]);
49
50      return 0;
51  }
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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